∑ Discrete Mathematics

GATE CSE 2027 · complete formula, rule, method & trap sheet

97 reference boxes · **9** chapters · **0** worked examples (switch on above)

Logic | Sets, Relations & Functions | Partial Orders & Lattices | Monoids & Groups | Combinatorics | Recurrence Relations & Generating Functions | Graph Theory | Problem → Standard Method Guide | High-Yield Extensions & Standard Methods

1 Logic

MUST REMEMBER

Implication

`P → Q ≡ ¬P ∨ Q`. It is false only when P is true and Q is false.

FORMULA / RULE

Equivalences

`P ↔ Q ≡ (P→Q) ∧ (Q→P)` ; De Morgan: `¬(P∧Q)=¬P∨¬Q` , `¬(P∨Q)=¬P∧¬Q` .

COMMON TRAP

Converse vs contrapositive

Contrapositive of `P→Q` is `¬Q→¬P` and is equivalent. Converse `Q→P` is generally not.

GATE FAVOURITE

Quantifiers

Negation flips quantifier: `¬∀x P(x) ≡ ∃x ¬P(x)` , `¬∃x P(x) ≡ ∀x ¬P(x)` .

FORMULA / RULE

Rules of inference (valid argument forms)

Rule	Form
Modus ponens	`P, P→Q ⊢ Q`
Modus tollens	`¬Q, P→Q ⊢ ¬P`
Hypothetical syllogism	`P→Q, Q→R ⊢ P→R`
Disjunctive syllogism	`P∨Q, ¬P ⊢ Q`
Resolution	`P∨Q, ¬P∨R ⊢ Q∨R`
Simplification / Addition	`P∧Q ⊢ P` · `P ⊢ P∨Q`

An argument is **valid** iff the premises true can never coexist with a false conclusion, i.e. (premises ∧ ¬conclusion) is unsatisfiable.

COMMON TRAP

Two fallacies that look like valid rules

Affirming the conclusion: `P→Q, Q ⊢ P` is **invalid**. **Denying the hypothesis:** `P→Q, ¬P ⊢ ¬Q` is **invalid**. Counter-example for both: P = F, Q = T makes the premises true and the conclusion false.

MUST REMEMBER

Normal forms & counting

CNF = AND of ORs of literals; **DNF** = OR of ANDs. A formula in CNF is **valid** iff every clause contains some `x` and `¬x` ; a formula in DNF is **satisfiable** iff some term has no clashing literals. n variables → `2ⁿ` truth-table rows and `2^(2ⁿ)` different Boolean functions (n = 2 gives 16).

GATE FAVOURITE

Translating English into FOL; free vs bound variables

All S are L: `∀x (S(x)→L(x))` . **Some S are L:** `∃x (S(x)∧L(x))` . Using ∧ with ∀ (or → with ∃) is the classic wrong answer. **Order matters:** `∀x∃y Loves(x,y)` = “everybody loves somebody”, while `∃y∀x Loves(x,y)` = “somebody is loved by everybody” (much stronger). A variable is **bound** if a quantifier covers it, otherwise **free**; a formula with no free variables is a **sentence** and has a truth value.

KEY POINT

Is the argument valid?

Premises: if it rains the match is cancelled; the match is not cancelled.
Conclusion: it did not rain.

Let R = rain, C = cancelled: `R→C, ¬C ⊢ ¬R` . This is **modus tollens**, so the argument is valid. Had the second premise been “the match is cancelled”, the argument would commit the fallacy of affirming the conclusion and be invalid.

FORMULA / RULE

Functional completeness

A set of connectives is **functionally complete** if every Boolean function can be built from it.

Set	Status	Why
{∧, ¬}, {∨, ¬}, {→, ¬}	Complete	De Morgan gives the missing one: p∨q = ¬(¬p∧¬q); p∨q = ¬p→q.
{↑} (NAND) alone, {↓} (NOR) alone	Complete	¬p = p↑p; p∧q = ¬(p↑q). NAND and NOR are the only single binary connectives that are complete.
{∧, ∨}	Not complete	Only monotone functions — no way to produce ¬p.
{→} alone	Not complete	Always 1 on the all-true row, so ¬p (false on that row) is unreachable.
{⊕, ∧}, {⊕, ¬}, {↔, ¬}	Not complete	{⊕,∧} keeps 0→0 so it never makes ¬p or the constant 1; {⊕,¬} gives only affine functions. {⊕, ∧, constant 1} *is* complete.

Post's test: a set is complete iff it is *not* contained in any one of the classes 0-preserving, 1-preserving, monotone, self-dual, affine. With n variables there are 2^{2^n} Boolean functions and $2^{2^{n-1}}$ self-dual ones (16 for n = 3).

GATE FAVOURITE

Quantifier distribution — which direction is valid?

∀ distributes over ∧ and ∃ over ∨ **exactly**: ∀x(P∧Q) ≡ ∀xP ∧ ∀xQ, ∃x(P∨Q) ≡ ∃xP ∨ ∃xQ. For the other pairings only **one direction** holds:

Formula	Status	Reason
∀xP ∨ ∀xQ → ∀x(P∨Q)	Valid	Converse is false
∃x(P∧Q) → ∃xP ∧ ∃xQ	Valid	Converse is false
(∀xP → ∀xQ) → ∀x(P→Q)	Not valid	Make ∀xP false, and P true / Q false at some x
∃y∀x R(x,y) → ∀x∃y R(x,y)	Valid	Converse is false (swap of quantifiers changes meaning)

Counter-model for the invalid ones: domain {1, 2}, P true only at 1, Q true only at 2 — then ∀x(P∨Q) is true but ∀xP ∨ ∀xQ is false.

MUST REMEMBER

Valid, satisfiable, unsatisfiable

Valid (tautology) = true under every interpretation. **Satisfiable** = true under at least one. **Unsatisfiable** (contradiction) = true under none. A is valid ⇔ ¬A is unsatisfiable. A formula can be satisfiable yet not valid (e.g. P → Q). Propositional SAT is NP-complete; validity in first-order logic is undecidable (but semi-decidable).

MUST REMEMBER

Free and bound variables

An occurrence of x is **bound** if it lies inside the scope of a quantifier ∀x or ∃x; otherwise it is **free**. A quantifier's scope is the formula that immediately follows it. A formula with **no free variables** is a **sentence** (closed formula) and is simply true or false in a given domain; one with free variables is true or false only after the variables are given values. The same letter can be free in one place and bound in another, and a quantified variable can be renamed (`∃x Q(x) ≡ ∃z Q(z)`) as long as the new name does not clash with a free variable.

Worked example. In `∀x (P(x, y) → ∃z Q(x, z)) ∧ R(x)` , the ∀x covers only `P(x, y) → ∃z Q(x, z)` , so x is bound there. The final `R(x)` lies outside that scope, so that x is **free**; y is free (nothing quantifies it); z is bound by ∃z. Free variables: **y and the x in R(x)**, so the formula is not a sentence.

2 Sets, Relations & Functions

MUST REMEMBER

Relation properties

Reflexive: (a,a) for every a. Symmetric: aRb ⇒ bRa. Antisymmetric: aRb and bRa ⇒ a=b. Transitive: aRb and bRc ⇒ aRc.

GATE FAVOURITE

Equivalence relation

Exactly: reflexive + symmetric + transitive. Equivalence classes form a partition of the set.

FORMULA / RULE

Function composition

If `f:A→B` and `g:B→C` , then `g∘f:A→C` . Composition order matters.

COMMON TRAP

Injective/surjective

Injective concerns distinct inputs having distinct outputs. Surjective means every codomain element is hit. Neither implies the other for finite sets of unequal size.

FORMULA / RULE

Counting relations and functions (A has n elements; A→B uses |A|=m, |B|=n)

Object	Count
All relations on A	`2^(n²)`
Reflexive / irreflexive	`2^(n²−n)` each
Symmetric	`2^(n(n+1)/2)`
Antisymmetric	`2ⁿ · 3^(n(n−1)/2)`
Asymmetric	`3^(n(n−1)/2)`
Equivalence relations	Bell number: 1, 2, 5, 15, 52, 203 for n = 1…6
Functions A→B / injective / bijective	`nᵐ` / `P(n,m)` / `m!` (when m = n)
Onto functions A→B	`Σ (−1)ᵏ C(n,k)(n−k)ᵐ = n!·S(m,n)`

Check with n = 3: antisymmetric = $2^3 \cdot 3^3 = 216$; asymmetric = 27; equivalence = 5.

MUST REMEMBER

Closures of a relation

Reflexive closure = R ∪ Δ. **Symmetric closure** = R ∪ R^{-1}. **Transitive closure** R^+ = smallest transitive relation containing R (Warshall's algorithm, $O(n^3)$). Equivalence closure = take reflexive, symmetric and transitive closure; apply the transitive step **last**. Example: R = {(1,2),(2,3),(3,4)} has transitive closure with **6** pairs: the three originals plus (1,3), (2,4), (1,4).

GATE FAVOURITE

Properties of functions and composition

For finite sets of equal size, injective ⇔ surjective ⇔ bijective. If `g∘f` is injective then **f** is injective; if `g∘f` is surjective then **g** is surjective. A function has an inverse iff it is bijective. `|P(A)| = 2ⁿ` , `|A×B| = |A||B|` .

KEY POINT

How many equivalence relations on a 4-element set have exactly 2 classes?

Count partitions into 2 blocks: Stirling number `S(4,2) = 7` (four of type 3+1 and three of type 2+2). Adding the other block counts: `S(4,1)=1` , `S(4,3)=6` , `S(4,4)=1` totals the Bell number **15**.

MUST REMEMBER

Cardinality and countability

Countably infinite (same size as ℕ): ℤ, ℚ, ℕ×ℕ, the set of all finite strings over a finite alphabet, any finite union or product of countable sets, and any subset of a countable set. **Uncountable**: ℝ, the power set $\mathcal{P}(\mathbb{N})$, the set of all infinite binary strings, the set of all functions ℕ→{0,1} (Cantor's diagonal argument). For any set A, $|\mathcal{P}(A)| = 2^{|A|} > |A|$. Programs (or Turing machines) are finite strings, hence countable, while languages are uncountable — so most languages are not recognised by any machine.

FORMULA / RULE

Composition, inverse and extra counts for relations

Composition (apply R first, then S): `S∘R = {(a,c) : ∃b, aRb and bSc}` ; `(S∘R)⁻¹ = R⁻¹∘S⁻¹` . With Boolean matrices $M_{S\circ R} = M_R \cdot M_S$ (Boolean product). R is transitive iff $R \circ R \subseteq R$; the transitive closure is $R \cup R^2 \cup \ldots \cup R^n$. On an n-element set: reflexive **and** antisymmetric = $3^{n(n-1)/2}$ (27 for n = 3); symmetric **and** antisymmetric = 2^n (subsets of the diagonal); partial orders on 3 labelled elements = 19; transitive relations on 3 elements = 171. Idempotent functions ($f \circ f = f$) on n points: $\Sigma_k\, C(n,k) \cdot k^{n-k}$ (n = 3 gives 3 + 6 + 1 = 10).

COMMON TRAP

Unions and intersections of equivalence relations

The **intersection** of equivalence relations is an equivalence relation, but the **union** need not be: {1~2} ∪ {2~3} lacks (1,3), so transitivity fails. The smallest equivalence relation containing both is the transitive closure of the union.

3 Partial Orders & Lattices

MUST REMEMBER

Partial order

A relation that is reflexive, antisymmetric and transitive. A poset is `(P,≤)`

GATE FAVOURITE

Hasse diagram

Represent the cover relation only; omit reflexive and transitive edges. Read minimal/maximal elements from the diagram carefully.

COMMON TRAP

Minimal vs least

A least element is ≤ every element and is unique if it exists. A poset may have multiple minimal elements without having a least element.

SHORTCUT

Finite lattice check

Every pair must have both a unique least upper bound (join) and a unique greatest lower bound (meet).

MUST REMEMBER

Bounds, chains and antichains

An **upper bound** of a subset S is an element ≥ every member of S; the **least upper bound** (join) is the smallest such bound, and the **greatest lower bound** (meet) is defined dually. A **chain** is a set of pairwise comparable elements and an **antichain** a set of pairwise incomparable ones. **Dilworth:** the minimum number of chains needed to cover a finite poset equals the size of its largest antichain.

FORMULA / RULE

Facts about the divisor lattice D_n ($n = p_1^{a_1} \cdot p_2^{a_2} \cdot \ldots$)

Elements: $\prod(a_i+1)$ ($360 = 2^3 \cdot 3^2 \cdot 5$ has 24 divisors). Meet = gcd, join = lcm. D_n is always **distributive**; it is a **Boolean algebra iff n is square-free**, and then it has 2^k elements for k primes. **Hasse-diagram edges** = $\Sigma_i\, a_i \cdot \prod_{j \neq i}(a_j+1)$: D12 gives 2·2 + 1·3 = 7, D36 gives 2·3 + 2·3 = 12, D30 gives 12 (a cube).

COMMON TRAP

Lattice types

Bounded = has top and bottom (every finite lattice is). **Complemented** = every element has a complement ($x \wedge y = \bot$ and $x \vee y = \top$). **Distributive** = $x \wedge (y \vee z) = (x \wedge y) \vee (x \wedge z)$. A lattice is distributive iff it contains no copy of the diamond M_3 or pentagon N_5. A **Boolean algebra** is a complemented distributive lattice, so it has 2^k elements. A complemented lattice need not be distributive.

KEY POINT

Is the poset ({1,2,3,6,12,18}, divisibility) a lattice?

Check pairs that might fail: lcm(12,18) = 36, which is not in the set, so 12 and 18 have **no upper bound at all**. Hence **not a lattice**. A common mistake is to test only gcd; both meet and join must exist for every pair.

FORMULA / RULE

Chains, antichains and Dilworth in divisor and subset posets

Dilworth: minimum number of chains covering a poset = size of the largest antichain; **Mirsky** (dual): minimum number of antichains = length of the longest chain. In D_n with $n = p_1^{a_1} \cdot p_2^{a_2} \cdot \ldots$: longest chain has $\Sigma a_i + 1$ elements; the number of **maximal chains** = $(\Sigma a_i)! / \prod a_i!$ (D_{36}: 4!/(2!2!) = 6; D_{360}: 6!/(3!2!1!) = 60). In the subset lattice $\mathcal{P}(n)$: n! maximal chains and largest antichain $C(n, \lfloor n/2 \rfloor)$ (Sperner). For divisibility on {1,…,12} the largest antichain is {7,…,12} (size 6), so 6 chains suffice and are needed.

COMMON TRAP

Distributive, complemented and the forbidden sublattices

A lattice is distributive iff it has no sublattice isomorphic to the diamond **M_3** or the pentagon **N_5** (N_5: $0 < a < c < 1$ and $0 < b < 1$; there $c \wedge (a \vee b) = c$ but $(c \wedge a) \vee (c \wedge b) = a$). Both are complemented lattices that are **not** distributive; in N_5 the element b has two complements (a and c). In a distributive lattice complements are unique, so Boolean algebras have unique complements. A chain with more than two elements is distributive but not complemented.

4 Monoids & Groups

MUST REMEMBER

Hierarchy

Semigroup = associative. Monoid = associative + identity. Group = monoid + inverse for every element.

GATE FAVOURITE

Group facts

Identity is unique; inverse of each element is unique. In a finite group, cancellation holds.

COMMON TRAP

Closure is separate

Associativity alone does not make a structure a semigroup. Verify closure under the operation too.

FORMULA / RULE

Algebraic structures at a glance

Structure	Needs	Examples
Semigroup	closure + associativity	$(\mathbb{N}^+, +)$, $(\mathbb{Z}, \times)$
Monoid	semigroup + identity	$(\mathbb{N}, +)$ with 0, $(\mathbb{Z}, \times)$
Group	monoid + inverses	$(\mathbb{Z}, +)$, $(\mathbb{Q} \setminus \{0\}, \times)$, Z_n under + mod n
Abelian group	group + commutative	$(\mathbb{Z}, +)$
Ring / Field	two operations, distributive / every nonzero element has a ×-inverse	Z_n is a field iff n is prime

$(\mathbb{Z}, -)$ is **not** a semigroup because subtraction is not associative.

MUST REMEMBER

Cyclic groups (order n, generator g)

The number of generators is $\varphi(n)$. There is exactly **one subgroup per divisor** of n. The number of elements of order d (for d | n) is $\varphi(d)$. In Z_{12}: 4 generators (1, 5, 7, 11); the elements of order 4 are 3 and 9 ($\varphi(4) = 2$). The order of a in Z_n is $n / \gcd(a, n)$.

GATE FAVOURITE

Lagrange and its consequences

|H| divides |G| for every subgroup H, so the order of every element divides |G|; a group of **prime order is cyclic**; the converse of Lagrange is false (A_4 has order 12 but no subgroup of order 6). If $a^2 = e$ for all a then G is abelian; every group of even order has an element of order 2. Groups of order 4: Z_4 and the Klein four-group; order 6: Z_6 and S_3.

SHORTCUT

Subgroup test

A non-empty $H \subseteq G$ is a subgroup iff $a, b \in H \Rightarrow a \cdot b^{-1} \in H$. For a **finite** H it is enough that H is closed under the operation.

MUST REMEMBER

Permutation groups

S_n has n! elements and is non-abelian for $n \geq 3$. Write permutations as disjoint cycles: the **order** is the lcm of the cycle lengths, and f^k only depends on k mod order (for (1 2 3)(4 5), $f^{100} = f^4 = (1\ 2\ 3)$). A k-cycle is even iff k is odd; parity is multiplicative; A_n (even permutations) has n!/2 elements and is normal in S_n. Elements with cycle type having m_k cycles of length k: $n! / \prod(k^{m_k} \cdot m_k!)$. In S_5 the orders are 1,2,3,4,5,6 with max 6 (type 3+2, 20 elements); in S_4 there are 8 elements of order 3; S_7 has max order 12 (type 4+3). A_4 has order 12 but no subgroup of order 6.

FORMULA / RULE

Normal subgroups, homomorphisms and direct products

H is **normal** if $gHg^{-1} = H$ for all g; every subgroup of index 2 is normal and every kernel is normal. **First isomorphism theorem:** $G/\ker f \cong \text{im } f$, so $|\text{im } f| = |G|/|\ker f|$. Number of homomorphisms $Z_m \to Z_n = \gcd(m, n)$ (f(1) must have order dividing m). $Z_m \times Z_n$ is cyclic iff $\gcd(m,n) = 1$; the order of (a,b) in a product is lcm(|a|,|b|). Groups of order 4: Z_4, Klein V_4; order 6: Z_6, S_3; order 8: five groups.

FORMULA / RULE

The group of units Z_n^*

$Z_n^* = \{a : \gcd(a,n) = 1\}$ under multiplication mod n has φ(n) elements. It is **cyclic iff n = 1, 2, 4, p^k or $2p^k$** (p an odd prime). So Z_9^*, Z_{10}^*, Z_{18}^* are cyclic, while Z_8^* and Z_{12}^* are the Klein four-group and $Z_{15}^* \cong Z_2 \times Z_4$ (order 8, no element of order 8) are not.

5 Combinatorics

FORMULA / RULE

Permutations & combinations

`nPr = n!/(n-r)!`, `nCr = n!/[r!(n-r)!]`, and `nCr=nC(n-r)`.

GATE FAVOURITE

Inclusion–Exclusion

For two sets: `|AUB|=|A|+|B|-|A∩B|`. For many sets, alternate signs across intersections.

SHORTCUT

At least one

Often easier as **total − none**. For independent repeated choices, compute the complement first when the event is “at least one”.

COMMON TRAP

Pigeonhole

If n objects are placed into k boxes and n>k, at least one box has ≥2 objects. General form: some box has at least `ceil(n/k)` objects.

FORMULA / RULE

Stars and bars (distributing identical objects)

Problem	Count
$x_1+\ldots+x_k = n$, all $x_i \geq 0$	`C(n+k−1, k−1)`
All $x_i \geq 1$	`C(n−1, k−1)`
$x_i \geq a_i$ (lower bounds)	substitute $y_i = x_i - a_i$, then use the first row with $n - \Sigma a_i$
Upper bounds $x_i \leq u$	inclusion–exclusion on the variables that break the bound

Example: $x_1+x_2+x_3+x_4 = 20$ with every $x_i \geq 2$ → shift to sum 12 → `C(15,3) = 455`

FORMULA / RULE

Binomial theorem and identities

`(x+y)^n = Σ C(n,k) x^(n−k) y^k`. Coefficient of x^3 in $(2+x)^5$ = `C(5,3)·2² = 40`. `ΣC(n,k) = 2^n`; alternating sum = 0; Pascal: `C(n,k) = C(n−1,k−1)+C(n−1,k)`; Vandermonde: `Σ C(m,i)C(n,r−i) = C(m+n,r)` (e.g. 35 = C(7,3)); hockey stick: `Σ_{i=r}^{n} C(i,r) = C(n+1,r+1)`. Multinomial: arrangements = `n!/(n1!n2!…)`.

MUST REMEMBER

Special counting sequences

Sequence	Values	Counts
Derangements D_n	0, 1, 2, 9, 44, 265 (n = 1…6)	permutations with no fixed point; `D_n = (n−1)(D_{n−1}+D_{n−2})`, ≈ n!/e
Catalan C_n = C(2n,n)/(n+1)	1, 1, 2, 5, 14, 42, 132	binary search trees on n keys, balanced bracket strings, full binary trees with n+1 leaves
Stirling S(n,k)	S(4,2)=7, S(5,3)=25	partitions of n objects into k non-empty blocks
Bell B_n	1, 2, 5, 15, 52	all partitions of an n-set = Σ_k S(n,k)

KEY POINT

5 men and 3 women in a row, no two women adjacent

Seat the men first: 5! = 120. They create 6 gaps; choose 3 for the women: C(6,3) = 20; arrange the women: 3! = 6. Total = 120 × 20 × 6 = **14400**.

FORMULA / RULE

Circular arrangements and restricted arrangements

n distinct objects in a circle: (n − 1)!. Two particular objects adjacent: 2·(n − 2)!; not adjacent: (n − 1)! − 2·(n − 2)!. m men and m women alternating: (m − 1)! ·m!. Bracelet (flip also allowed): (n − 1)!/2. In a line, use the **block method** (glue items that must be together) or the **gap method**: arrange the unrestricted items first, then choose gaps for the restricted ones — r items that may not touch go into C(k + 1, r) of the k + 1 gaps around k other items.

Example: MISSISSIPPI with no two S adjacent: arrange M,I,I,I,I,P,P in 7!/(4!2!) = 105 ways, then pick 4 of the 8 gaps: 105 × C(8,4) = 105 × 70 = **7350**.

FORMULA / RULE

Distribution problems at a glance

Situation	Count
n distinct objects → k distinct boxes	k^n
n identical objects → k distinct boxes (empty allowed / none empty)	C(n+k−1, k−1) / C(n−1, k−1)
n distinct objects → k identical boxes, none empty	Stirling S(n, k)
n identical objects → identical boxes (integer partitions)	p(n): p(6) = 11, p(8) = 22
Lattice paths in an m × n grid	C(m+n, m); not crossing the diagonal: Catalan C_n

COMMON TRAP

Pigeonhole variants

To guarantee r objects in one of k boxes you need (r − 1)·k + 1 objects (37 people ⇒ 4 share a birth month). Among any n + 1 integers two leave the same remainder mod n. **Divisibility form:** from {1,…,2n} any n + 1 numbers contain two with one dividing the other — group the numbers by their odd part (there are n groups), because two numbers with the same odd part divide each other. The bound is tight: {n+1,…,2n} has no such pair.

6 Recurrence Relations & Generating Functions

MUST REMEMBER

Linear homogeneous recurrence

For constant coefficients, form the characteristic equation. Repeated roots require polynomial factors in n.

FORMULA / RULE

Repeated root

Root r of multiplicity k contributes `(C0+C1n+...+C_{k−1}n^(k−1))r^n`

GATE FAVOURITE

Particular solution

For non-homogeneous recurrences, choose an appropriate particular form; if it overlaps a homogeneous root, multiply the trial by the required power of n.

SHORTCUT

Generating functions

Translate sequence terms into a power series, manipulate algebraically, then extract the coefficient. Keep indexing consistent.

KEY POINT

Non-homogeneous with an overlapping root

`a_n = 5a_{n−1} − 6a_{n−2} + 2^n`, $a_0 = 0$, $a_1 = 1$. Homogeneous roots are 2 and 3. The forcing term 2^n overlaps root 2, so the trial particular solution is `C·n·2^n` (multiply by n). Substituting gives C = −2. General solution: `A·2^n + B·3^n − 2n·2^n`; a_0 and a_1 give A = −5, B = 5, so $a_n = 5\cdot3^n - 5\cdot2^n - 2n\cdot2^n$ (check: $a_2 = 45 - 20 - 16 = 9 = 5\cdot1 - 0 + 4$).

FORMULA / RULE

Standard generating functions

Sequence	Generating function
1, 1, 1, …	`1/(1−x)`
1, a, a², …	`1/(1−ax)`
C(n+k−1, k−1) (k-fold repetition)	`1/(1−x)^k`
C(n, r)	`(1+x)^n`
1/r!	`e^x`

Rules: a product of generating functions is the convolution of the sequences; multiplying by `1/(1−x)` gives partial sums; shifting by x^m moves the index.

KEY POINT

Counting with a generating function

Ways to pay n using 1-coins and 2-coins: coefficient of x^n in `1/((1−x)(1−x²))` = ⌊n/2⌋ + 1, so n = 10 gives **6**. Another standard model: tilings of a 2×n board with dominoes follow Fibonacci (1, 2, 3, 5, 8 …), so a 2×5 board has **8** tilings.

FORMULA / RULE

Trial form for the particular solution

Forcing term	Trial solution
constant c	A (multiply by n if 1 is a characteristic root)
polynomial of degree d in n	polynomial of degree d (× n^m if 1 is a root of multiplicity m)
$c \cdot r^n$	$A \cdot r^n$ (× n^m if r is a root of multiplicity m)
(polynomial)$\cdot r^n$	(polynomial of the same degree)$\cdot r^n$

Then add the homogeneous solution and fit the initial conditions. Check by iterating the first few terms.

FORMULA / RULE

Standard results worth memorising

`a_n = a_{n-1} + n, a_0 = 0` ⇒ n(n+1)/2. `a_n = 2a_{n-1} + n, a_0 = 1` ⇒ $3 \cdot 2^n - n - 2$ ($a_4 = 42$). `T(n) = 2T(n/2) + n, T(1) = 1` ⇒ $n \cdot \log_2 n + n$ (T(16) = 80). Climbing stairs with 1, 2 or 3 steps: tribonacci, 1, 2, 4, 7, 13, 24, 44, 81, 149, **274** for 10 steps. Ternary strings with no two adjacent 0s: $a_n = 2a_{n-1} + 2a_{n-2}$ (1, 3, 8, 22, 60, 164, 448, 1224). Regions made by n lines in general position: 1 + n(n+1)/2.

SHORTCUT

Generating functions: coins, bounds and parity

Coins of 1, 2, 5: `1/((1−x)(1−x²)(1−x⁵))` — the coefficient of x^n is the number of ways to pay n (n = 15 gives 18). A variable with $0 \le x \le u$ contributes $(1 - x^{u+1})/(1 - x)$; an even count contributes $1/(1 - x^2)$ and an odd count $x/(1 - x^2)$. Partitions of n into parts of size at most k: $\prod_{i \le k} 1/(1 - x^i)$. Without a generating function, a quick hand-count: fix the number of 5-coins, then count (1,2)-ways as ⌊r/2⌋ + 1.

7 Graph Theory

MUST REMEMBER

Handshake lemma

`Σ deg(v)=2|E|` for undirected graphs. Therefore the number of odd-degree vertices is even.

GATE FAVOURITE

Connectivity

For an undirected graph, connected means every pair of vertices has a path. A tree is connected and acyclic.

FORMULA / RULE

Tree invariant

A tree with n vertices has exactly `n−1` edges. Adding one edge creates exactly one cycle; deleting any edge disconnects the tree.

COMMON TRAP

Euler vs Hamilton

Euler concerns using every **edge**; Hamilton concerns visiting every **vertex**. Do not interchange their conditions.

FORMULA / RULE

Euler and Hamilton: the exact tests

Euler circuit: connected and every degree even (directed: in-degree = out-degree). **Euler path:** connected with 0 or 2 odd-degree vertices. K_n has an Euler circuit iff n is odd. **Hamilton:** no simple iff-test; **Dirac** ($n \ge 3$, every degree ≥ n/2) and **Ore** (deg u + deg v ≥ n for non-adjacent pairs) are sufficient only. K_n is Hamiltonian for $n \ge 3$; $K_{m,n}$ is Hamiltonian iff $m = n \ge 2$.

FORMULA / RULE

Planarity and Euler's formula

Connected planar graph: `V − E + F = 2`. For $V \ge 3$: `E ≤ 3V − 6`; if there is no triangle (e.g. bipartite): `E ≤ 2V − 4`. Sum of face degrees = 2E. K_5 (10 edges > 9) and $K_{3,3}$ (9 edges > 8) are non-planar, and by Kuratowski a graph is planar iff it has no subdivision of either. Example: a connected planar cubic graph on 8 vertices has E = 12 and F = 2 − 8 + 12 = **6**.

FORMULA / RULE

Spanning trees

Cayley: K_n has `n^(n−2)` spanning trees ($K_4 \to 16$). $K_{m,n}$ has `m^(n−1)·n^(m−1)` ($K_{2,3} \to 12$; $K_{3,3} \to 81$). The cycle C_n has n. In general use the matrix-tree theorem (any cofactor of the Laplacian). A spanning tree has n−1 edges; if all edge weights are distinct the minimum spanning tree is unique (Kruskal/Prim); cut property and cycle property justify both algorithms.

MUST REMEMBER

Connectivity and special graph counts

Vertex connectivity κ ≤ edge connectivity λ ≤ minimum degree δ. In a tree every edge is a bridge and every non-leaf is a cut vertex. A forest with n vertices and c components has n − c edges. K_n has n(n−1)/2 edges; $K_{m,n}$ has mn. There are `2^(n(n−1)/2)` labelled simple graphs on n vertices (64 for n = 4) but only 11 non-isomorphic ones. A graph is bipartite iff it has no odd cycle.

MUST REMEMBER

Directed graphs

Σ in-degree = Σ out-degree = |E|. The (i,j) entry of A^k counts walks of length k from i to j. A digraph is **strongly connected** if every vertex reaches every other; a **DAG** has a topological order (a vertex with in-degree 0 always exists).

MUST REMEMBER

Matching, vertex cover, independent set, edge cover

For a graph on n vertices: **α + τ = n** (independent set + vertex cover) and **ν + ρ = n** (matching + edge cover, no isolated vertices — Gallai). **König (bipartite graphs):** ν = τ, hence α = n − ν; Hall's condition characterises perfect matchings. Perfect matchings: $K_{n,n}$ has n!, K_{2n} has (2n − 1)!! (K_6: 15, K_8: 105); a tree has at most one; every regular bipartite graph has one. The Petersen graph has a perfect matching and independence number 4.

MUST REMEMBER

Graph colouring

$\chi(G) \geq \omega(G)$ (clique number) but not always equal: the Grötzsch graph is triangle-free with $\chi = 4$, and C_5 has $\omega = 2$, $\chi = 3$. Brooks: $\chi \leq \Delta$ unless G is complete or an odd cycle; $\chi \leq \Delta + 1$ always; planar $\Rightarrow \chi \leq 4$; bipartite $\Leftrightarrow \chi \leq 2$; the Petersen graph has $\chi = 3$. **Chromatic polynomial** P(G,k) (colourings with k colours): tree $k(k-1)^{n-1}$; K_n $k(k-1)\ldots(k-n+1)$; cycle C_n $(k-1)^n + (-1)^n(k-1)$; deletion–contraction $P(G) = P(G - e) - P(G / e)$; disconnected graphs multiply. **Edge colouring:** $\Delta \leq \chi' \leq \Delta + 1$ (Vizing); bipartite $\chi' = \Delta$; $\chi'(K_n) = n - 1$ for even n and n for odd n.

FORMULA / RULE

Degree sequences, complements and extremal counts

Havel–Hakimi: sort descending, delete the largest d_1 and subtract 1 from the next d_1 entries; repeat. The sequence is graphic iff you reach all zeros without a negative entry. An even sum is necessary, not sufficient. Complement: $|E(G)| + |E(\bar{G})| = C(n,2)$; a self-complementary graph has $n \equiv 0$ or $1 \pmod 4$ (P_4, C_5). Mantel: a triangle-free graph has at most $\lfloor n^2/4 \rfloor$ edges (attained by $K_{\lfloor n/2 \rfloor, \lceil n/2 \rceil}$). Unlabelled trees on n = 1…8 vertices: 1, 1, 1, 2, 3, 6, 11, 23; non-isomorphic simple graphs on n = 1…6: 1, 2, 4, 11, 34, 156; connected 3-regular graphs on 6 vertices: 2 ($K_{3,3}$ and the prism).

FORMULA / RULE

Rooted trees and m-ary trees

A full m-ary tree with i internal vertices has $n = m \cdot i + 1$ vertices and $(m - 1) \cdot i + 1$ leaves (binary: i + 1 leaves, 2i + 1 vertices). A binary tree of height h (edges) has at most $2^{(h+1)} - 1$ vertices and at most 2^h leaves; with n_2 vertices of degree 2 it has $n_2 + 1$ leaves. Binary search trees on n keys: Catalan number C_n. A tree has one or two centre vertices.

FORMULA / RULE

Connectivity, planarity bounds, Euler and Hamilton extras

A **cut vertex** increases the number of components when removed; a **bridge** is an edge with that property; **blocks** are the maximal 2-connected pieces, and consecutive blocks meet at cut vertices. Planar with girth g: $E \leq g(V - 2)/(g - 2)$: triangle-free (g = 4) gives $E \leq 2V - 4$ ($K_{2,8}$ reaches 16 for V = 10), and the Petersen graph (g = 5, $E = 15 > 5 \cdot 8/3$) is non-planar. $K_{m,n}$ has an Euler circuit iff m and n are both even; it has an Euler trail but no circuit when exactly two vertices have odd degree ($K_{2,3}$). The Petersen graph has a Hamilton path but **no** Hamilton cycle.

8 Problem → Standard Method Guide

METHOD

Logic & Proof Patterns

Truth/equivalence question → use truth table or standard logical equivalences. Check only the relevant rows/identities.

Implication validity → rewrite `P→Q` as `¬PvQ` ; look for the single false case.

Quantifier negation → push negation inward while flipping $\forall \leftrightarrow \exists$. Do not negate the predicate incorrectly.

Argument validity → translate premises/conclusion into logic; test whether premises true can coexist with conclusion false.

METHOD

Relations, Functions, Posets & Lattices

Classify a relation → systematically test reflexive, symmetric, antisymmetric, transitive using counterexamples where possible.

Equivalence classes → verify equivalence relation, then group elements related to a chosen representative.

Partial-order question → check R, AS, T. For Hasse diagrams, remove loops and transitive edges.

Least/greatest/minimal/maximal → compare using the order relation, not visual position alone.

Lattice question → for each relevant pair, find both meet (GLB) and join (LUB).

Function classification → injective: compare inputs/outputs; surjective: check every codomain value has a preimage; bijective: both.

METHOD

Combinatorics

"At least one" → complement counting: total − none.

Overlapping cases → Inclusion–Exclusion; watch alternating signs.

Repeated objects / identical items → divide by factorials of repetition counts when appropriate.

Restricted arrangements → count unrestricted arrangements and subtract invalid cases, or use block/gap methods depending on the restriction.

Guaranteed repetition → Pigeonhole principle; formulate the boxes and objects explicitly before calculating.

Choose r from n → combinations when order does not matter; permutations when it does.

METHOD

Recurrences & Generating Functions

Constant-coefficient linear recurrence → characteristic equation → roots → homogeneous solution → particular solution → initial conditions.

Repeated characteristic root → multiply successive terms by powers of n.

Non-homogeneous recurrence → solve homogeneous part first, then construct a particular solution.

Generating-function problem → define the series, transform the recurrence/sequence algebraically, then extract coefficients.

METHOD

Extended GATE Calculation Patterns

Problem signal	Standard method	Key check / shortcut
CNF/DNF conversion	Use logical equivalences; for canonical forms enumerate assignments if needed.	CNF is AND of OR clauses; DNF is OR of AND terms.
Relation represented as matrix	Use rows/columns to test reflexivity, symmetry, antisymmetry; Boolean matrix multiplication can test composition/transitivity.	Transitivity requires $R \circ R \subseteq R$.
Counting onto functions	Inclusion–Exclusion over missing codomain values.	For $n \to m$, onto is possible only when $n \geq m$.
Arrangements with identical items	Multiset permutation formula.	Divide by factorials of repeated multiplicities.
Restricted circular arrangement	Fix a reference position when rotations are considered identical.	Do not divide by n!; circular permutations usually use (n−1)! for distinct objects.
Recurrence order / characteristic roots	Write characteristic polynomial, factor it, apply multiplicity rule.	Initial conditions determine constants, not the characteristic roots.
Recurrence with polynomial/exponential forcing	Guess a particular form matching the forcing term.	If the guess overlaps the homogeneous solution, multiply by n until independent.
Generating-function coefficient	Convert to known series or partial fractions, then extract coefficient.	Index shifts carefully; write first few terms to verify.
Maximum/minimum degree bound	Handshake lemma + graph constraints.	Odd-degree vertices must occur in an even count.
Connected graph edge bounds	Use tree as minimum-edge connected graph and complete graph as maximum simple graph.	Connected simple graph: $V-1 \leq E \leq V(V-1)/2$.
Euler circuit/trail	Use degree parity + connectivity of non-isolated vertices.	All even → Euler circuit; exactly two odd → Euler trail but not circuit.
Bipartite graph	2-colouring via BFS/DFS or identify an odd cycle.	No odd cycle ⇔ bipartite.
Matching question	Look for augmenting paths; use structural constraints/bipartite matching reasoning.	A matching never shares a vertex between selected edges.
Graph colouring lower bound	Find a clique or odd-cycle obstruction; then construct a colouring for an upper bound.	$\chi(G) \geq \omega(G)$; bipartite nonempty graph has $\chi=2$.
Group identity/inverse	Test closure, associativity, identity and inverses under the given operation.	Identity and inverse are unique in a group.
Subgroup test	For a nonempty subset H, verify closure under a^{-1}b for all a,b∈H.	This single test replaces separate closure + inverse checks.
Finite group order	Use divisibility properties of subgroup/order when applicable.	Order of an element divides the order of a finite group.
Which connective sets are complete?	Show each of ¬, ∧ (or ∨) is expressible; otherwise use a class that is closed (monotone, 0- or 1-preserving, affine, self-dual).	$\{\wedge, \vee\}$ and $\{\to\}$ alone are not complete; NAND and NOR alone are.
Is a quantified implication valid?	Try a counter-model on a 2-element domain: P true at one point, Q true at the other.	∃ with ∧ and ∀ with ∨ only give one direction.
Closure / composition of a relation	Boolean-matrix product or Warshall; count pairs of every reachability component (a cycle of length k contributes k^2 pairs).	Compose in the stated order; closure on a cycle is complete.
Minimum chains / maximum antichain	Dilworth: find the largest set of pairwise incomparable elements.	For divisibility on {1..2n}: {n+1..2n} is an antichain of size n.
Order of a permutation / power of it	Disjoint cycles, then lcm of cycle lengths; reduce the exponent modulo the order.	Cycle type 3+2 has order 6; a k-cycle has parity $k-1$.
Is Z_n^* cyclic?	Cyclic iff $n = 1, 2, 4, p^k, 2p^k$.	Z_8^*, Z_{12}^*, Z_{15}^* are not cyclic.
Homomorphisms $Z_m \to Z_n$	$f(1) = a$ with $m \cdot a \equiv 0 \pmod n$; count = gcd(m, n).	|image| = |G| / |kernel|.
Circular / forbidden-adjacency counting	Fix one person; treat adjacent pairs as a block; use the gap method for non-adjacent.	Not adjacent = $(n-1)! - 2(n-2)!$.
Pigeonhole with divisibility or sums	Choose boxes that force the property (odd part, remainder class).	Guarantee r in one box needs $(r-1)k + 1$ objects.
Non-homogeneous recurrence with overlap	Trial = $(n^m)\cdot$(forcing-term form); substitute to find the constant.	Verify by iterating three or four terms.
Coin-change / restricted-sum counting	Product of generating functions, or fix the largest coin and count the rest.	Bounded variable: $(1 - x^{u+1})/(1 - x)$.
Is a degree sequence graphic?	Havel–Hakimi reduction; an odd sum fails immediately.	Even sum is necessary, not sufficient.
Max matching / min vertex cover / max independent set	Find a maximum matching; in bipartite graphs $\tau = \nu$ (König) and $\alpha = n - \nu$.	A Hall violator shows no perfect matching.
Number of proper colourings	Deletion–contraction, or use the formula for trees, cycles and complete graphs.	C_n: $(k-1)^n + (-1)^n(k-1)$.
Cut vertices and bridges	Remove each vertex or edge and count components, or use low-link values.	A leaf is never a cut vertex; every tree edge is a bridge.
Edge bound for planar graphs	Triangle-free: $E \leq 2V - 4$; otherwise $E \leq 3V - 6$; girth g: $E \leq g(V-2)/(g-2)$.	Faces from Euler: $F = 2 - V + E$.

METHOD

Graph Theory

Degree-count problem → handshake lemma `Σdeg=2E`.

Tree problem → immediately use `E=V−1`, connected + acyclic, and unique simple path properties.

Bipartite test → 2-color via BFS/DFS; equivalently, no odd cycle.

Euler trail/circuit → reason from vertex degrees and connectivity of the non-isolated portion.

Matching → identify maximum-cardinality constraints and augmenting-path logic where applicable.

Graph colouring → use lower bounds (cliques/odd cycles), upper bounds, and structural properties rather than blindly trying colours.

9 High-Yield Extensions & Standard Methods

MUST REMEMBER

Graph isomorphism

An isomorphism is a bijection preserving adjacency. Before attempting a mapping, compare invariants: number of vertices/edges, degree multiset, connected components, cycle structure and counts of special vertices.

GATE FAVOURITE

Cosets

For subgroup H of finite group G, every left coset has |H| elements and cosets partition G. By Lagrange, |H| divides |G| and the number of distinct cosets is |G|/|H|.

FORMULA / RULE

Homomorphism

A map preserves the operation: `f(a*b)=f(a)*f(b)` . For group homomorphisms, the kernel is a subgroup and the image is a subgroup; injective iff the kernel is the identity-only set.

COMMON TRAP

Homomorphism vs isomorphism

A homomorphism only has to preserve the operation, so it need not be injective or onto; an isomorphism is a **bijective** homomorphism. The kernel measures failure of injectivity: for finite groups `|image| = |G| / |kernel|` .

MUST REMEMBER

Hall's condition

A bipartite graph has a matching saturating the left side iff for every subset S of left vertices, `|N(S)| ≥ |S|`

FORMULA / RULE

Colouring bounds

`χ(G) ≥ ω(G)` , where ω is clique number. A nonempty bipartite graph has chromatic number 2. For a complete graph K_n, χ=n.

MUST REMEMBER

Generating-function workflow

Translate the recurrence/counting condition into coefficient extraction. Use geometric-series identities for unrestricted repetition and multiply generating functions when independent choices combine.

SHORTCUT

Graph invariant filter

If two graphs have different degree multisets, component counts, or vertex/edge counts, they are immediately non-isomorphic; do not attempt a full mapping.

Representative worked problems

METHOD

Q1 · Cosets and subgroup order

Given: |G|=24 and H is a subgroup of order 6. **How many distinct left cosets of H exist?**

Method: Use the coset partition theorem: number of cosets = |G|/|H|.

`24/6 = 4` . Every coset contains exactly 6 elements and the four cosets partition G.

METHOD

Q2 · Graph isomorphism filter

Given: G has degree sequence (3,2,2,1,1,1) while H has (3,3,1,1,1,1). **Can they be isomorphic?**

Method: Isomorphism preserves the degree of every vertex, hence preserves the sorted degree multiset.

The sequences differ, so the graphs are **not isomorphic**. No vertex mapping needs to be attempted.

METHOD

Q3 · Hall's theorem

Given: Left vertices {a,b,c} have neighborhoods N(a)={1,2}, N(b)={2,3}, N(c)={1,2,3}. **Does a matching saturating the left side exist?**

Method: Check Hall's condition for every relevant subset. Singletons have at least one neighbor; {a,b} has {1,2,3}; {a,c} and {b,c} also have 3; all three have 3.

Thus `|N(S)|≥|S|` for every S, so Hall's theorem guarantees a matching saturating {a,b,c}. One such matching is a→1, b→2, c→3.

METHOD

Q4 · Generating function coefficient

Problem: Find the number of solutions to x+y+z=5 with x,y,z≥0.

Method: Each variable contributes `1/(1-t)` ; multiply and extract $[t^5]$.

`(1-t)^-3 = Σ C(n+2,2)t^n` , so the coefficient of t^5 is `C(7,2)=21` .